5.11 Riemann Surfaces

I develop the elementary theory of Riemann surface, starting with the more standard concept approach of maximal atlases (see [War13]) and then relating to the notion sheaves (see [Wed16] (Section 4.2)). The language of sheaves is well suited to complex analytic spaces and aligns well with the approach in the algebraic case.

Definition 5.11.1
*Let X be a topological space. By definition a **chart** on X is a pair (U, φ) where $U \subset X$ is an open set and φ is a homeomorphism onto an open subset $\varphi(U)$ of $\mathbb{C}$.*

*We say that two charts (U, φ) and (V, ψ) are **compatible** if either $U \cap V = \emptyset$ or the composite map $(\psi \circ \varphi^{-1}) : \varphi(U \cap V) \to \psi(U \cap V)$ is **biholomorphic**.*

*An **atlas** $\mathcal{A}$ on X is a collection of charts $\{(U_\alpha, \varphi_\alpha)\}$ for which $X = \bigcup_\alpha U_\alpha$ and every pair of charts is compatible.*

*We say two atlases $\mathcal{A}, \mathcal{A}'$ are **equivalent** if $\mathcal{A} \cup \mathcal{A}'$ is an atlas.*

Proposition 5.11.2
The notion of compatible charts is an equivalence relation. Similarly for the notion of equivalent atlases.

A union of equivalent atlases is again an atlas, and every atlas is contained in a unique maximal atlas.

We may then use the charts to define what it means for a function on a manifold to be holomorphic.

Lemma 5.11.3
Let X be a topological space, $U \subset X$ an open subset, $f : U \to \mathbb{C}$ a function and (V, φ), (V', ψ) two compatible charts such that $U \cap V \cap V' \neq \emptyset$.

Then $f \circ \varphi^{-1}|_{\varphi(U \cap V \cap V')}$ is holomorphic if and only if $f \circ \psi^{-1}|_{\psi(U \cap V \cap V')}$ is holomorphic.

Lemma 5.11.4
Let X be a topological space, $U \subset X$ an open subset, $f : U \to \mathbb{C}$ a function and $(V_\alpha, \varphi_\alpha)_{\alpha \in \mathcal{A}}$ a family of charts such that $U \subset \bigcup_{\alpha \in \mathcal{A}} V_\alpha$. Further suppose that (V, φ) is a chart such that $U \cap V \neq \emptyset$ and is compatible with every $(V_\alpha, \varphi_\alpha)$. Then the following are equivalent

a) $f \circ \varphi^{-1}|_{\varphi(V \cap U)}$ is holomorphic
b) $f \circ \varphi_\alpha^{-1}|_{\varphi_\alpha(V_\alpha \cap U \cap V)}$ is holomorphic for all $\alpha \in \mathcal{A}$ for which $V_\alpha \cap V \neq \emptyset$

Proposition 5.11.5 (Holomorphic Function)
Let X be a topological space, $U \subset X$ an open subset, $f : U \to \mathbb{C}$ a function and $\mathcal{A}$ a given atlas. Then the following are equivalent

a) there exists a collection of charts $\{(V_\alpha, \varphi_\alpha)\} \subset \mathcal{A}$ such that $V_\alpha \cap U \neq \emptyset$, $f \circ \varphi_\alpha^{-1}|_{\varphi_\alpha(V_\alpha \cap U)}$ is holomorphic and $U \subset \bigcup_\alpha V_\alpha$
b) for every chart $(V, \varphi) \in \mathcal{A}$ such that $U \cap V \neq \emptyset$ the function $f \circ \varphi^{-1}|_{\varphi(U \cap V)}$ is holomorphic.

*When this holds we say that f is **holomorphic** with respect to $\mathcal{A}$.*

Proof. $b) \implies a)$ Straightforward

$a) \implies b)$ As every chart (V, φ) is compatible with the collection $\{(V_\alpha, \varphi_\alpha)\}$ this follows from (5.11.4). ∎

Proposition 5.11.6
A holomorphic function $f : U \to \mathbb{C}$ is continuous.

Proof. For every chart (V, φ) the function $f \circ \varphi^{-1}|_{\varphi(U \cap V)}$ is holomorphic (5.11.5) and therefore continuous. Further by definition the map $\varphi|_{U \cap V}$ is continuous whence $f|_{U \cap V}$ is continuous (4.1.28). As the charts cover X we are done by (4.1.29). ∎

We first prove that this definition does not depend on which atlas is chosen.

Proposition 5.11.7
Let X be a topological space, $U \subset X$ an open subset, $f : U \to \mathbb{C}$ a function and $\mathcal{A}, \mathcal{A}'$ two equivalent atlases.

Then f is holomorphic with respect to $\mathcal{A}$ if and only if it is holomorphic with respect to $\mathcal{A}'$.

Proof. Since equivalence of atlases is symmetric we only need to show one direction. So suppose that f is holomorphic with respect to $\mathcal{A}$, and let $(W, \psi) \in \mathcal{A}'$ be a chart. Then by (5.11.4) $f \circ \psi^{-1}$ is holomorphic, so that f is holomorphic with respect to $\mathcal{A}'$. ∎

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Further we show that just as in the usual case this is a local condition.

Proposition 5.11.8 (Structure Sheaf associated to an Atlas)
Let X be a topological space together with an atlas $\mathcal{A}$. Let $U = \bigcup_{i\in I} U_i$ an open cover and $f : U \to \mathbb{C}$ a function. Then f is holomorphic (with respect to $\mathcal{A}$) if and only if $f|_{U_i}$ is holomorphic for all $i \in I$.

In particular for every atlas the mapping

$$\mathcal{O}_X(U) \coloneqq \{f : U \to \mathbb{C} \mid f \text{ holomorphic } \}$$

*determines a sheaf of $\mathbb{C}$-algebras, and the pair $(X, \mathcal{O}_X)$ constitutes a space of $\mathbb{C}$-functions, which we call the **structure sheaf**. It contains an isomorphic copy of $\mathbb{C}$ consisting of the constant functions.*

Two atlases are equivalent if and only if they determine the same structure sheaf.

A space of functions $(X, \mathcal{O}_X)$ arises in this way if and only if there is an open cover $X = \bigcup_{i\in I} U_i$ such that $\left(U_i, \mathcal{O}_X|_{U_i}\right) \xrightarrow{\sim} \left(V_i, \mathcal{O}_{V_i}\right)$ for $V_i \subset \mathbb{C}$ an open subset with the usual sheaf of holomorphic functions.

Proof. Suppose that $(V, \varphi) \in \mathcal{A}$ is a chart such that $V \cap U_i \neq \emptyset$. Then by assumption $f \circ \varphi^{-1}|_{\varphi(V\cap U)}$ is holomorphic whence by (...) $f|_{U_i} \circ \varphi^{-1}|_{\varphi(V\cap U_i)}$ is holomorphic. Since the chart (V, φ) was arbitrary we conclude $f|_{U_i}$ is holomorphic with respect to the atlas $\mathcal{A}$.

Conversely suppose $f|_{U_i}$ is holomorphic with respect to $\mathcal{A}$ for all $i \in I$. Consider an arbitrary chart $(V, \varphi) \in \mathcal{A}$ such that $V \cap U \neq \emptyset$. Then $V \cap U = \bigcup_{i\in I} V \cap U_i$, where we may restrict to open subsets with $V \cap U_i \neq \emptyset$. Then by assumption $f|_{U_i} \circ \varphi^{-1}|_{\varphi(V\cap U_i)} = \left(f \circ \varphi^{-1}|_{\varphi(V\cap U)}\right)|_{\varphi(V\cap U_i)}$ is holomorphic. By (...) $f \circ \varphi^{-1}|_{\varphi(V\cap U)}$ is holomorphic. As the chart was arbitrary we conclude that f is holomorphic with respect to $\mathcal{A}$.

The holomorphic functions form a $\mathbb{C}$-algebra by (5.5.4), and in particular contain the constant functions. Further $D(f) = X \setminus f^{-1}(\{0\})$ which is an open subset of X, and $\left(f|_{D(f)}\right)^{-1}$ is holomorphic whenever f is not identically zero.

By definition then $(X, \mathcal{O}_X)$ is a space of $\mathbb{C}$-functions. By (5.11.7) an equivalent atlas determines the same structure sheaf. Conversely suppose $\mathcal{A}, \mathcal{A}'$ determine the same structure sheaves. Consider two atlases $(V, \varphi) \in \mathcal{A}$ and $(V', \varphi') \in \mathcal{A}'$. Then by definition we see that $\varphi \circ (\varphi')^{-1}$ is holomorphic and $(\varphi') \circ \varphi^{-1}$ is holomorphic, and therefore they are compatible as required. ∎

Definition 5.11.9 (Riemann Surface)
Let $(X, \mathcal{O}_X)$ be a space of $\mathbb{C}$-functions, which is Hausdorff and second countable. Further suppose there is an open cover $X = \bigcup_{i\in I} U_i$ such that there exists isormorphisms

$$\varphi_i : \left(U_i, \mathcal{O}_X|_{U_i}\right) \xrightarrow{\sim} \left(V_i, \mathcal{O}_{V_i}\right)$$

*for $V_i \subset \mathbb{C}$ an open subset. Then we say X is a **Riemann Surface**.*

We say $\mathcal{A}$ is a compatible atlas if it determines the same structure sheaf. Further there exists a unique compatible maximal atlas (5.11.8).

Then $\varphi : U \to \mathbb{C}$ is a chart for X if it lies in at least one compatible atlas (equivalently is in the unique maximal atlas).

5.11.1 Morphisms of Riemann Surfaces

Proposition 5.11.10 (Holomorphic maps of Riemann Surfaces)
Let $f : X \to Y$ be a continuous map of Riemann surfaces with given compatible atlases $\mathcal{A}_1, \mathcal{A}_2$. Then the following are equivalent

a) $g \in \mathcal{O}_Y(V) \implies g \circ f \in \mathcal{O}_X(f^{-1}(V))$
b) For all charts $(V, \psi) \in \mathcal{A}_2$ the function $\psi \circ f|_{f^{-1}(V)}$ is holomorphic
c) For all charts $(U, \varphi) \in \mathcal{A}_1$, $(V, \psi) \in \mathcal{A}_2$ for which $U \cap f^{-1}(V) \neq \emptyset$ we have

$$\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \to \mathbb{C}$$

is holomorphic

*This definition is independent of the choice of atlases, and we say in this case that f is **holomorphic**.*

Proof. *a)* $\implies$ *c)* As $\psi \in \mathcal{O}_Y(V)$ is trivially holomorphic we have by assumption $\psi \circ f \in \mathcal{O}_X(f^{-1}(V)) \implies (\psi \circ f)|_{U\cap f^{-1}(V)} \in \mathcal{O}_X(U \cap f^{-1}(V))$. By definition this means $\psi \circ f \circ \varphi^{-1}$ is holomorphic.

c) $\implies$ *b)* By (5.11.5) $\psi \circ f|_{f^{-1}(V)}$ is holomorphic

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b) $\Longrightarrow$ *a)* Let $(V_\alpha, \psi_\alpha) \in \mathcal{A}_2$ be charts for which $V \subset \bigcup_\alpha V_\alpha$ and $V_\alpha \cap V \neq \emptyset$. Then

$$(g \circ f)|_{f^{-1}(V_\alpha)} = (g \circ \psi_\alpha^{-1}) \circ (\psi_\alpha \circ f)|_{f^{-1}(V_\alpha)}$$

is by definition holomorphic. Since this is a local condition (5.11.8) we conclude that $g \circ f$ is holomorphic as required.

The last statement follows because *a)* is independent of the choice of compatible atlases (5.11.8). ∎

5.11.2 Tangent and Cotangent Space

Definition 5.11.11 (Local Ring)
Let X be a Riemann surface and $x \in X$ a point. Then define the local ring

$$\mathcal{O}_{X,x} \coloneqq \varinjlim_{x \in U} \mathcal{O}_X(U)$$

to be the $\mathbb{C}$-algebra of germs at x. Every germ $\boldsymbol{f}$ has a well-defined value at x which we denote by $\boldsymbol{f}(x)$. The unique maximal ideal $\mathfrak{m}_{X,x}$ consists of those germs which vanish at x. Further the residue field $\mathbb{C}(\mathfrak{m}_{X,x})$ is isomorphic to $\mathbb{C}$ (that is, $\mathbb{C} \to \mathcal{O}_{X,x} \to \mathcal{O}_{X,x}/\mathfrak{m}_{X,x}$ is an isomorphism).

Proposition 5.11.12 (Cotangent Space is Finite-Dimensional)
The local ring $\mathcal{O}_{X,x} = \mathbb{C} \oplus \mathfrak{m}_{X,x}$ and there is a canonical isomorphism

$$\mathrm{Der}(\mathcal{O}_{X,x}, \mathbb{C}) \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{C}}(\mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2, \mathbb{C})$$

*For every $f \in \mathcal{O}_X(U)$ and $x \in U$ denote by $\mathrm{d}f(x)$ the image of $(f - f(x))$ in the **cotangent space** $T_x^\star X \coloneqq \mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2$.*

This has dimension 1, more explicitly for any open chart (U, φ) around x, $0 \neq \mathrm{d}\varphi(x)$ is a basis for $\mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2$.

We denote the inverse image of the dual basis $(\mathrm{d}\varphi(x))^\vee$ by $\left(\frac{\partial}{\partial \varphi}\right)|_x$ (which is therefore a basis for $\mathrm{Der}(\mathcal{O}_{X,x}, \mathbb{C})$).

Proof. Every germ $\boldsymbol{f}$ may be written as $\boldsymbol{f} = \boldsymbol{f}(x) + (\boldsymbol{f} - \boldsymbol{f}(x))$. This is unique decomposition by considering evaluation at x, which gives the first statement. The isomorphism follows the same lines as were proved for affine algebras that tangent space is dual to cotangent space (see (3.32.50)). Explicitly the map is given by $D(\mathrm{d}f(x)) \coloneqq Df_x$.

Let (U, φ) be a chart and define the holomorphic function $z(x') \coloneqq \varphi(x') - \varphi(x)$. For every $f \in \mathcal{O}_X(U)$ define $\hat{f} \coloneqq f \circ \varphi^{-1}$. Then by Taylor's theorem (5.9.29)

$$\hat{f}(z') = \hat{f}(\varphi(x)) + (z' - \varphi(x))\hat{f}'(\varphi(x)) + (z' - \varphi(x))^2 R_2(z')$$

where $R_2(z')$ is a holomorphic function on $\varphi(U)$. Therefore

$$f(x') = f(x) + z(x')(f \circ \varphi^{-1})'(\varphi(x)) + z(x')^2 (R_2 \circ \varphi)(x')$$

where $R_2 \circ \varphi$ is a holomorphic function on U. Therefore if $f(x) = 0$ we have $f_x - (f \circ \varphi^{-1})'(\varphi(x)) z_x \in \mathfrak{m}_{X,x}^2$ and $\mathrm{d}f(x) = (f \circ \varphi^{-1})'(x)\,\mathrm{d}z(x)$. In other words $\mathrm{d}z(x) = \mathrm{d}\varphi(x)$ is spanning.

Observe that $\hat{z}'(x) = 1$. We claim that for any f such that $f_x \in \mathfrak{m}_{X,x}^2$ we have $\hat{f}'(x) = 0$, which will show that $z_x \notin \mathfrak{m}_{X,x}^2$ and therefore $\mathrm{d}z(x) = \mathrm{d}\varphi(x) \neq 0$.

For in this case $\hat{f} = \sum_{i=1}^n \hat{g}_i \hat{h}_i$ when restricted to some open neighbourhood V, with $\hat{g}_i(\varphi(x)) = \hat{h}_i(\varphi(x)) = 0$. By the product rule $\hat{f}'(\varphi(x)) = 0$ as required. ∎

Definition 5.11.13
*Let X be a Riemann Surface and $x \in X$. We define the **tangent space** to be the $\mathbb{C}$-vector space of derivations*

$$T_x X \coloneqq \mathrm{Der}(\mathcal{O}_{X,x}, \mathbb{C})$$

By (5.11.12) it has dimension 1, more explicitly given any chart (U, φ) around x it has basis $\left(\frac{\partial}{\partial \varphi}\right)|_x$ the unique derivation such that

$$\left(\frac{\partial}{\partial \varphi}\right)|_x \, \varphi_x = 1$$

Lemma 5.11.14 (Derivations are Directional Derivatives)
Let $f \in \mathcal{O}_X(U)$, $x \in U$ and (V, φ) a chart around x, then

$$\left(\frac{\partial}{\partial \varphi}\right)|_x \, f_x = (f \circ \varphi^{-1})'(\varphi(x))$$

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In particular

$$\mathrm{d}f(x) = \left(f \circ \varphi^{-1}\right)'(\varphi(x))\,\mathrm{d}\varphi(x)$$

Proof. As the derivative of a function depends only on its germ, the right hand side is a well-defined derivation on $\mathcal{O}_{X,x}$, by the usual rules of differentiation (5.5.4). Further $(\varphi \circ \varphi^{-1})(z) = z$ which has derivative 1 and so both sides are equal when evaluated at $f_x = \varphi_x$. By uniqueness (5.11.13) then they are equal. ∎

Corollary 5.11.15 (Change of Variables Formula)
Let X be a Riemann Surface and (U, φ), (V, ψ) two charts around x. Then

$$\mathrm{d}\varphi(x) = \left(\varphi \circ \psi^{-1}\right)'(\psi(x))\,\mathrm{d}\psi(x)$$

and dually

$$\left(\frac{\partial}{\partial \psi}\right)|_x = \left(\varphi \circ \psi^{-1}\right)'(\psi(x))\left(\frac{\partial}{\partial \varphi}\right)|_x$$

Proof. By (5.11.12) both $\mathrm{d}\varphi(x)$ and $\mathrm{d}\psi(x)$ form a basis for $T_x^\star X$, whence $\mathrm{d}\varphi(x) = \lambda\,\mathrm{d}\psi(x)$ for some $\lambda \in K^\star$. Therefore $\varphi_x - \lambda\psi_x \in \mathbb{C} + \mathfrak{m}_{X,x}^2$. By (...)

$$0 = \left(\frac{\partial}{\partial \psi}\right)|_x\ (\varphi_x - \lambda\psi_x) = \left(\varphi \circ \psi^{-1}\right)'(\psi(x)) - \lambda$$

as required. ∎

5.11.3 Holomorphic 1-forms

Lemma 5.11.16
Let X be a Riemann surface, $U \subset X$ an open subset and $\omega : U \to \coprod_{x\in U} T_x^\star X$ a function satisfying $\omega(x) \in T_x^\star X$. Then for every chart (V, φ) such that $U \cap V \neq \emptyset$

$$\omega(x) = \omega_\varphi(x)\,\mathrm{d}\varphi(x) \quad \forall x \in U \cap V$$

$$\omega_\varphi(x) := \left(\frac{\partial}{\partial \varphi}\right)|_x\ \omega(x)$$

Suppose that (W, ψ) is another chart such that $U \cap V \cap W \neq \emptyset$ then

$$\omega_\psi(x) = \left(\varphi \circ \psi^{-1}\right)'(\psi(x))\omega_\varphi(x) \quad \forall x \in U \cap V \cap W$$

Definition 5.11.17 (Holomorphic 1-form)
*Let X be a Riemann surface, $U \subset X$ an open subset and $\mathcal{A}$ a compatible atlas. We say a function $\omega : U \to \coprod_{x\in U} T_x^\star X$ is a **holomorphic 1-form** if*

a) $\omega(x) \in T_x^\star X$ for all $x \in U$
b) for every chart $(V, \varphi) \in \mathcal{A}$ such that $U \cap V \neq \emptyset$, the function $\omega_\varphi : U \cap V \to \mathbb{C}$ is holomorphic

Lemma 5.11.18
The definition of holomorphic 1-form (5.11.17) is independent of the choice of atlas.

Proof. Suppose ω is a holomorphic 1-form on U with respect to the atlas $\mathcal{A}$. Let $(V_i, \varphi_i) \in \mathcal{A}$ and (W, ψ) be compatible charts such that $U \cap V_i \cap W \neq \emptyset$. Then by (5.11.16) and (5.11.15) for all $x \in U \cap V_i \cap W$

$$\omega_\psi(x) = \left(\varphi_i \circ \psi^{-1}\right)'(\psi(x))\omega_{\varphi_i}(x)$$

whence by definition $\omega_\psi \circ \varphi_i^{-1} : \varphi_i(U \cap V_i \cap W) \to \mathbb{C}$ is holomorphic. Then by (5.11.5) ω_ψ is holomorphic on $U \cap W$. Therefore if $\mathcal{A}'$ is a compatible atlas we find ω is holomorphic with respect to $\mathcal{A}'$. By symmetry we see that the definition is independent of the choice of atlas. ∎

Proposition 5.11.19
Let X be a Riemann surface and $f \in \mathcal{O}_X(U)$. Then the function

$$\begin{aligned} \mathrm{d}f : U &\longrightarrow \coprod_{x\in U} T_x^\star U \\ x &\longrightarrow \mathrm{d}f(x) \end{aligned}$$

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is a holomorphic 1*-form on* U*. Moreover for any chart* (U, φ) *we have*

$$\mathrm{d}f(x) = \left(\partial \frac{f}{\partial \varphi}\right)|_x \, \mathrm{d}\varphi(x) = \left(f \circ \varphi^{-1}\right)'(\varphi(x)) \, \mathrm{d}\varphi(x)$$

Proposition 5.11.20
Let X *be a Riemann surface. For every open subset* $U \subset X$ *define*

$$\Omega^1_X(U) \coloneqq \left\{ \omega : U \to \coprod_{x \in U} T^\star_x X \;\middle|\; \omega \text{ holomorphic } \ 1\text{-form} \right\}$$

Then Ω^1_X *is a locally-free* $\mathcal{O}_X$*-module of rank* 1*, more precisely for every chart* (U, φ) *we have* $\Omega^1_X|_U$ *has global basis* $\mathrm{d}\varphi|_U$*.*

Proof. Evidently $\Omega^1_X(U)$ is a vector space over $\mathbb{C}$, as $\mathcal{O}_X(U)$ forms a $\mathbb{C}$-algebra. Further for $f \in \mathcal{O}_X(U)$ we find $f\omega$ is a holomorphic 1-form and $(f\omega)|_V = f|_V \, \omega|_V$, so that Ω^1_X is an $\mathcal{O}_X$-module. For every chart (U, φ) and open subset $V \subset U$ we have an isomorphism of $\mathcal{O}_X(V)$-modules

$$\begin{aligned} \mathcal{O}_X(V) &\xrightarrow{\sim} \Omega^1_X(V) \\ f &\to f \, \mathrm{d}\varphi|_U \\ \omega_\varphi &\leftarrow \omega \end{aligned}$$

whence an isomorphism

$$\mathcal{O}_X|_U \xrightarrow{\sim} \Omega^1_X|_U$$

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